EECS 484, Winter 2026- Database Management Systems

Normalization

Chapter 19

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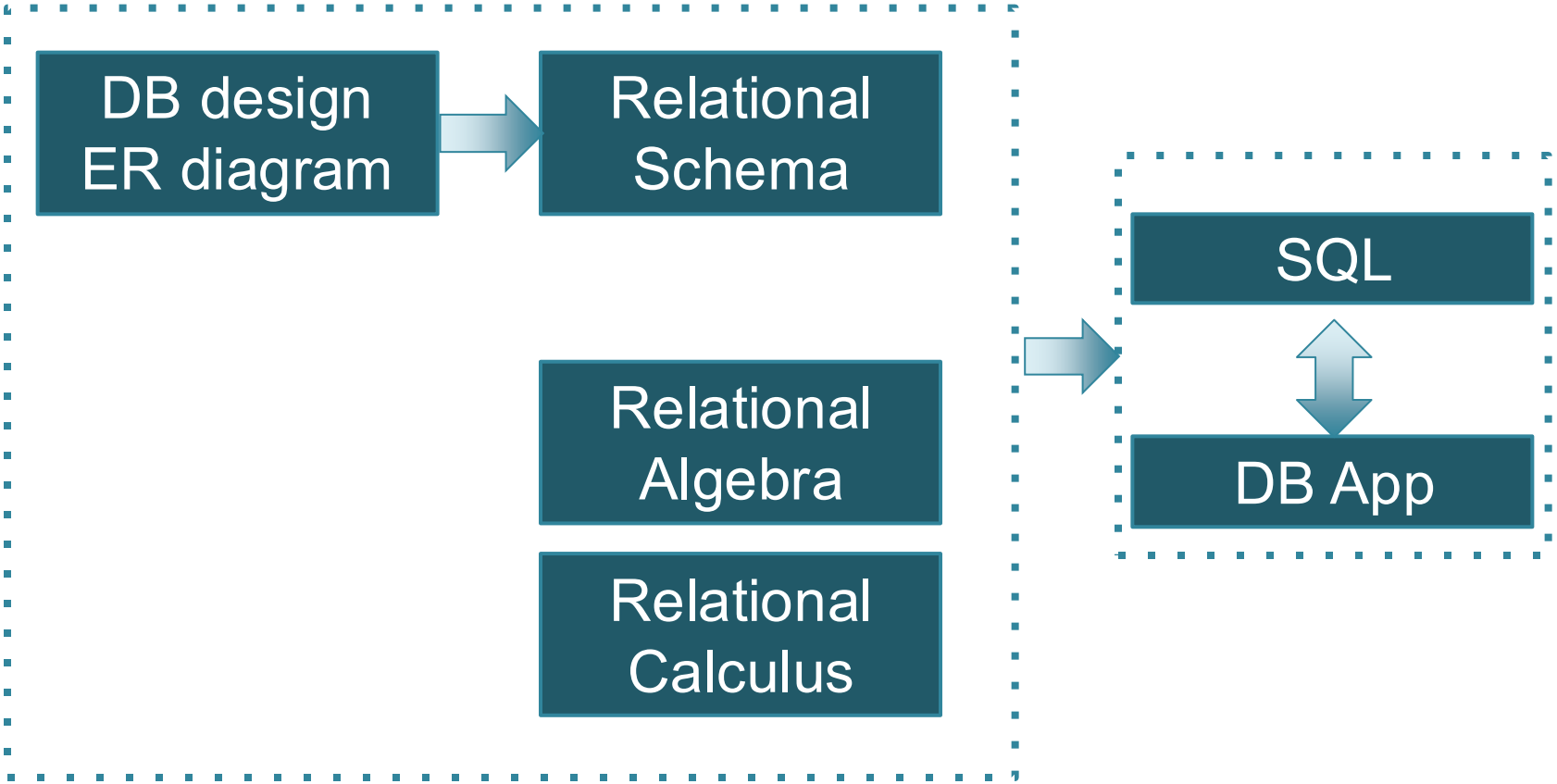
Review


DB design
ER diagram
Relational
Schema
Relational
Algebra
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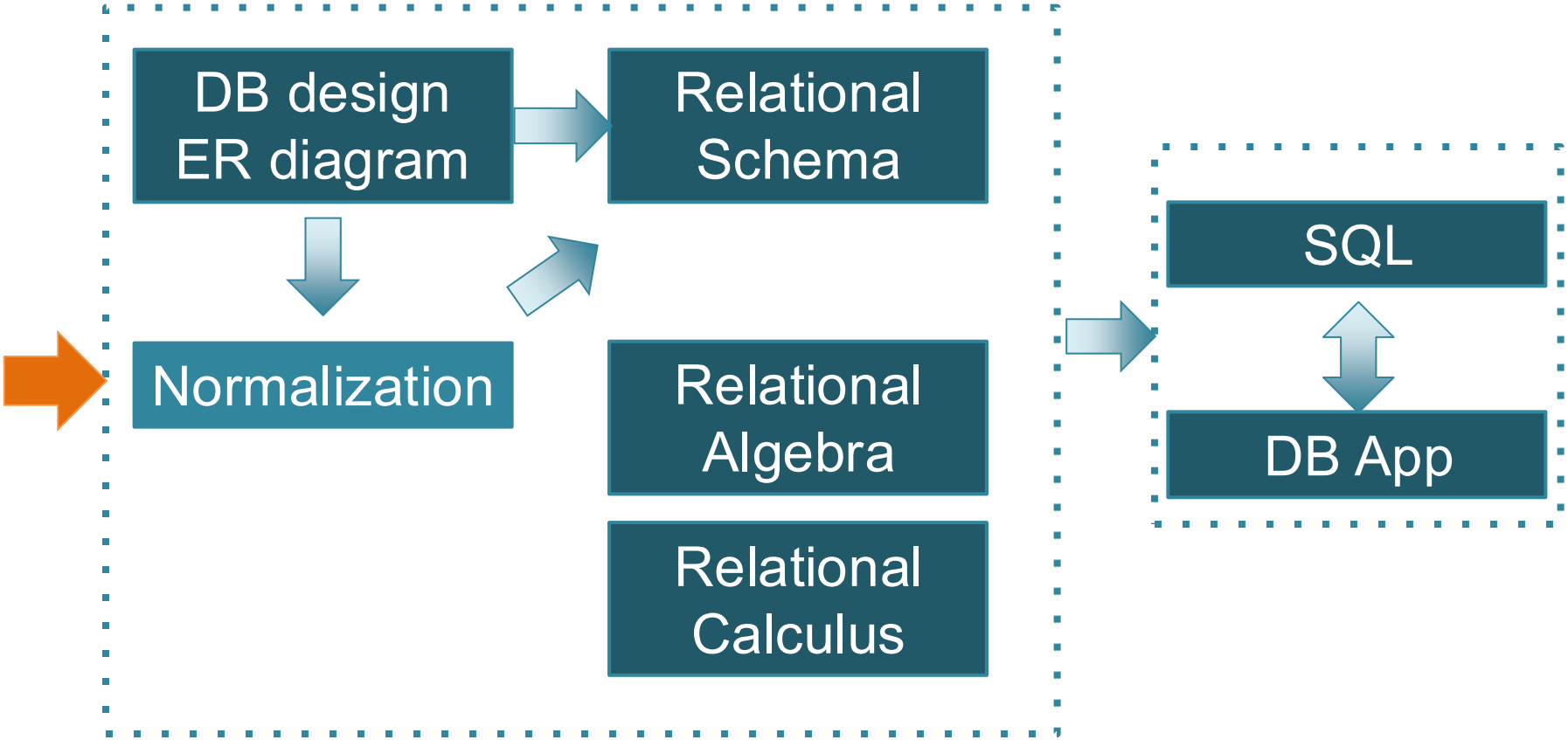
Calculus
SQL
DB App

Today


DB design
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Database Design: The Story So Far

- Requirements Analysis
  - Data stored, operations, apps, …
- Conceptual Database Design
  - Model high-level description of the data, constraints, ER model
- Logical Database Design
  - Choose a DBMS and design a database schema
- Schema Refinement
  - Normalize relations, avoid redundancy, anomalies…
- Physical Database Design
  - Examine physical database structures like indices, restructure…
- Security Design

Form/Spreadsheet

Supplier ID	Supplier Name	Supplier Address	Item	Desc	Price
1	Acme	A1	Dynamite	boom	$7
			Paint	blue	$10
			Flowers	pink	$3
2	Beanery	A2, A3	Dynamite	boom	$8

Good or bad table?

Form/Spreadsheet

Supplier ID	Supplier Name	Supplier Address	Item	Desc	Price
1	Acme	A1	Dynamite	boom	$8
			Paint	blue	$10
			Flowers	pink	$3
2	Beanery	A2, A3	Dynamite	boom	$8

Problems

- (Supplier ID, item) appears to be the key, but it is NULL in many places=> Implicit dependencies between rows
- Addresses can be multi-valued

Normalization

Supplier ID	Supplier Name	Supplier Address	Item	Desc	Price
1	Acme	A1	Dynamite	boom	$7
			Paint	blue	$10
			Flowers	pink	$3
2	Beanery	A2, A3	Dynamite	boom	$8

The above is not a good table!

Going to proper set of tables is called "normalization".

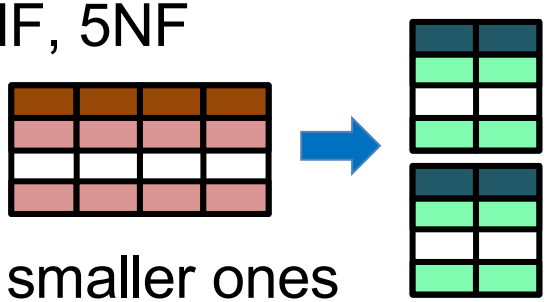
- avoid redundancy of data
- capture the dependencies inherent in the data

Normalization Theory

Goal: Design ‘good’ tables

Normalization is like doing laundry.
It feels good in theory, but nobody wants to do it all the way to BCNF.

- What is good?
  - We want tables where the attributes depend on the primary key, on the whole key, and nothing but the key.
- How to check for goodness?
  - Use normal forms (NFs): each NF excludes certain “anomalies”, 1NF, 2NF, 3NF, BCNF, 4NF, 5NF
- How to fix bad tables?
  - Decomposition: break larger tables into smaller ones



1st Normal Form – First Step

Supplier ID	Supplier Name	Supplier Address	Item	Desc	Price
1	Acme	A1	Dynamite	boom	$7
1	Acme	A1	Paint	blue	$10
1	Acme	A2	Flowers	pink	$3
2	Beanery	A2	Dynamite	boom	$8
2	Beanery	A3	Dynamite	boom	$8

1. Each ingle-valued
2. Each row contains all the relevant data
   - Exception: we allow NULLs, unless part of primary key

1 & 2 ⇒We have a relational table:

Rows can be reordered, all rows independent.

What is Wrong with Redundancy?

Supplier ID	Supplier Name	Supplier Address	Item	Desc	Price
1	Acme	A1	Dynamite	boom	$7
1	Acme	A1	Paint	blue	$10
1	Acme	A1	Flowers	pink	$3
2	Beanery	A2	Dynamite	boom	$8
2	Beanery	A3	Dynamite	boom	$8

Redundant storage: For each different item, we also store the address of the supplier.

Redundancy Problems

- Redundant storage (space)
  - A supplier supplies multiple items
- Update anomalies
  - Change address of a supplier
  - Need to change all instances!
- Insertion anomalies
  - Insert a supplier
  - Has to insert other (unrelated) info too
- Deletion anomalies
  - What if we want to delete the last item tuple?
  - Has to delete other (unrelated) info too

Supplier ID	Supplier Name	Supplier Address	Item	Desc	Price
1	Acme	A1	Dynamite	boom	$8
1	Acme	A1	Paint	blue	$10
1	Acme	A2	Flowers	pink	$3
2	Beanery	A2	Dynamite	boom	$8
2	Beanery	A[illegible]	[illegible]	[illegible]	[illegible]

Dealing with Redundancy

- Redundancy arises when schema forces an unnatural association among attributes
- The new trick we will learn today is to use the notion of functional dependencies

Solution to Redundancy: Decomposition

Split large relation to smaller ones

Supplier ID	Supplier Name	Supplier Address	Item	Desc	Price
1	Acme	A1	Dynamite	boom	$7
1	Acme	A1	Paint	blue	$10
1	Acme	A1	Flowers	pink	$3
2	Beanery	A2	Dynamite	boom	$8
2	Beanery	A3	Dynamite	boom	$8

Supplier ID	Supplier Name
1	Acme
2	Beanery

Item	Desc
Dynamite	boom
Paint	blue
Flowers	pink

Supplier ID	Supplier Address
1	A1
2	A2
2	A3

Supp ID	Item	Price
1	Dynamite	$7
1	Paint	$10
1	Flowers	$3
2	Dynamite	$8

- What columns to decompose on?
  - Need to learn about Normal Forms

Normal Forms

- Certain kind of decomposition
- Guarantees that certain problems won't occur & obeys certain rules:
  - 1 NF : No set-valued attrs
  - 2 NF : Historical
  - 3 NF : …
  - BCNF : Boyce-Codd Normal Form

Functional Dependencies (FD)

FDs capture dependencies among attributes

Supplier ID	Supplier Name	Supplier Address	Item	Desc	Price
1	Acme	A1	Dynamite	boom	$7
1	Acme	A1	Paint	blue	$10
1	Acme	A2	Flowers	pink	$3
2	Beanery	A2	Dynamite	boom	$8
2	Beanery	A3	Dynamite	boom	$8

- 'Suppl depends on the Supp
- What does 'depends on' mean?

FD: Definition

- Notation: a → b
- Read as: ‘a’ functionally determines ‘b’
- **Informally**: If you know ‘a’, there is only one ‘b’ to match.

Supplier ID	Supplier Name	Supplier Address	Item	Desc	Price
1	Acme	A1	Dynamite	boom	$7
1	Acme	A1	Paint	blue	$10
1	Acme	A2	Flowers	pink	$3
2	Beanery	A2	Dynamite	boom	$8
2	Beanery	A3	Dynamite	boom	$8

FD: Definition

- Notation: a → b
- Read as: ‘a’ functionally determines ‘b’
- **Informally**: If you know ‘a’, there is only one ‘b’ to match.
- **Formally**: A form of Integrity Constraint

D: $X \rightarrow Y$ X and Y subsets of relation R’s attributes

$t1 \in r,\ t2 \in r,\ \prod_X (t1) = \prod_X (t2) \Rightarrow \prod_y (t1) = \prod_y (t2)$

Supplier ID	Supplier Name	Supplier Address	Item	Desc	Price
1	Acme	A1	Dynamite	boom	$7
1	Acme	A1	Paint	blue	$10
1	Acme	A2	Flowers	pink	$3
2	Beanery	A2	Dynamite	boom	$8
2	Beanery	A3	Dynamite	boom	$8

FD: Example

FDs capture dependencies among attributes

Supplier ID	Supplier Name	Supplier Address	Item	Desc	Price
1	Acme	A1	Dynamite	boom	$7
1	Acme	A1	Paint	blue	$10
1	Acme	A2	Flowers	pink	$3
2	Beanery	A2	Dynamite	boom	$8
2	Beaner			boom	$8

FD: Example

FDs capture dependencies among attributes

Supplier ID	Supplier Name	Supplier Address	Item	Desc	Price
1	Acme	A1	Dynamite	boom	$7
1	Acme	A1	Paint	blue	$10
1	Acme	A2	Flowers	pink	$3
2	Beanery	A2	Dynamite	boom	$8
2	Beanery	A2	Dynamite	boom	$8

- Supplier ID → Supplier Name
- Item → Desc
- Supplier ID, Item → Price

More on FDs

- An FD is a statement about all allowable relations.
  - Based only on application semantics, not a table instance

Primary Key IC: special case of FD

- Role of FDs in detecting redundancy

Example: Relation R with 4 attributes, XYZK

X → Y causes redundancy

(X,Y)=(1,10) and (X,Y)=(2,20) stored multiple times

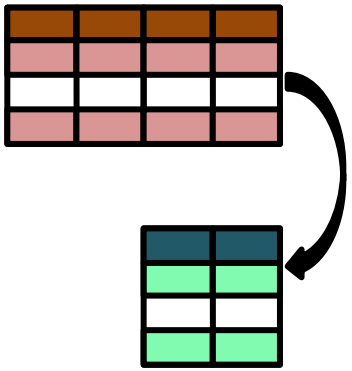
X	Y	Z	K
1	10	11	A
1	10	12	B
2	20	22	C
2	20	23	D

How to find all the implied FDs?

- F+: Closure of F = Set of all FDs (including the derived ones)

Supplier ID → Supplier Name
Item → Desc
Supplier ID, Item → Price
…

- How to obtain?
  - F+ obtained by repeatedly applying Armstrong's Axioms



Armstrong's Inference Axioms

- Axiom#1: Reflexive Property:
  - (Supplier ID, Item#) -> Item#
  - Obviously, if we know (Supplier ID and item#) pair, we know the item#
- In general, given attribute sets X and Y
  - Reflexivity: If $X \supseteq Y$, then $X \rightarrow Y$
- In the above example, Y is [Item#]. X is [Supplier ID, Item#].
- This is called a trivial dependency.

All the Inference Axioms

- Armstrong's Axioms (X, Y, Z are sets of attributes): 'sound' and 'complete'
  - Reflexivity: If $X \supseteq Y$, then $X \rightarrow Y$ (trivial dependency)
  - Augmentation: If $X \rightarrow Y$, then $XZ \rightarrow YZ$ for any Z
  - Transitivity: If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$

    e.g. ename → ejob, ejob → esal; ⇒ ename → esal
- Additional useful rules (derivable):
  - Union: If $X \rightarrow Y$ and $X \rightarrow Z$, then $X \rightarrow YZ$
  - Decomposition: If $X \rightarrow YZ$, then $X \rightarrow Y$ and $X \rightarrow Z$

Deriving Union Rule from Axioms

- Prove: if $X \rightarrow Y$ and $X \rightarrow Z$ then $X \rightarrow YZ$
- Proof:

- Reflexivity: If $Y \subseteq X$, then $X \rightarrow Y$ (trivial dependency)
- Augmentation: If $X \rightarrow Y$, then $XZ \rightarrow YZ$ for any Z
- Transitivity: If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$

Deriving Union Rule from Axioms

- Prove: if $X \rightarrow Y$ and $X \rightarrow Z$ then $X \rightarrow YZ$
- Proof:

  1. $X \rightarrow Y$ (given)
  2. $X \rightarrow Z$ (given)
  3. $XX \rightarrow XZ$ or $X \rightarrow XZ$ (augmentation of 2)
  4. $XZ \rightarrow YZ$ (augmentation of 1)
  5. $X \rightarrow YZ$ (transitivity of 3 and 4)

- Possible to derive the decomposition rule from the basic Armstrong rules

Solution to Redundancy: Decomposition

Split large relation to smaller ones

Supplier ID	Supplier Name	Supplier Address	Item	Desc	Price
1	Acme	A1	Dynamite	boom	$7
1	Acme	A1	Paint	blue	$10
1	Acme	A1	Flowers	pink	$3
2	Beanery	A2	Dynamite	boom	$8
2	Beanery	A3	Dynamite	boom	$8

Supplier ID	Supplier Name
1	Acme
2	Beanery

Item	Desc
Dynamite	boom
Paint	blue
Flowers	pink

Supplier ID	Supplier Address
1	A1
2	A2
2	A3

Supp ID	Item	Price
1	Dynamite	$7
1	Paint	$10
1	Flowers	$3
2	Dynamite	$8

Advantages:

- Eliminate redundancy and anomalies
- Lossless join when done right

Solution to Redundancy: Decomposition

- Split large relations to smaller ones

Supplier ID	Supplier Name	Supplier Address	Item	Desc	Price
1	Acme	A1	Dynamite	boom	$7
1	Acme	A1	Paint	blue	$10
1	Acme	A1	Flowers	pink	$3
2	Beanery	A2	Dynamite	boom	$8
2	Beanery	A3	Dynamite	boom	$8

Supplier ID	Supplier Name
1	Acme
2	Beanery

Supplier ID	Supplier Address
1	A1
2	A2
2	A3

Item	Desc
Dynamite	boom
Paint	blue
Flowers	pink

Supp ID	Item	Price
1	Dynamite	$7
1	Paint	$10
1	Flowers	$3
2	Dynamite	$8

- Problems with decomposition
  - Some queries become more expensive (more joins)
  - Lossless Join: Can we reconstruct the original relation from instances of the decomposed relations?

Must have

  - Dependency Preservation: Can we check dependencies without having to join the decomposed relations?

Good to have

Lossless Join Decompositions

- Relation R, FDs F; Decomposed to X, Y
- Lossless-Join decomposition if:

$$\pi_X(r) \bowtie \pi_Y(r) = r \quad \text{for } \textbf{every} \text{ instance } r \text{ of } R$$

- Note, $r \subseteq \pi_X(r) \bowtie \pi_Y(r)$ is always true, not vice versa, unless the join is lossless
- Can generalize to three or more relations

Example: R = {A, B, C}, X={A,B}, Y={B,C}

R

A	B	C
1	2	3
4	5	6
7	2	8

$\pi_X(R)$

A	B
1	2
4	5
7	2

$\pi_Y(R)$

B	C
2	3
5	6
2	8

$\pi_X(R) \bowtie \pi_Y(R)$

A	B	C
1	2	3
4	5	6
7	2	8
1	2	8
7	2	3

Lossless Join (cont.)

- Relation R, FDs F; Decomposed to X, Y
  - Test: lossless-join w.r.t. F if and only if F^+ contains:

$$X \cap Y \rightarrow X, \quad \text{or} \quad X \cap Y \rightarrow Y$$

i.e. attributes common to X and Y contain a super key for either X or Y

(Note: Different test needed for decomposition into more than two relations.)

Lossless join decomposition is always required!

Decomposition into 3+ relations

- There is an if-and-only-if test called **Chase Algorithm** (not covered in this class)
- **A practical shortcut:**
  - if you can show your N-way decomposition was produced by a sequence of binary splits, and each binary split satisfies the two-relation test ($X \cap Y \rightarrow X$ or $X \cap Y \rightarrow Y$), then the overall decomposition is guaranteed lossless.
  - Note: This shortcut is sufficient but not necessary: some lossless decompositions can't be verified this way and require the full chase.

Lossless Decomposition: Example

ssn	cid	grade	name	addr
123	413	A	Smith	Main
123	415	B	Smith	Main
234	211	A	Jones	Huron

ssn->name, address
ssn, cid -> grade

R1

ssn	cid	grade
123	413	A
123	415	B
234	211	A

ssn, cid -> grade

R2

ssn	name	addr
123	Smith	Main
234	Jones	Huron

ssn->name, address

Dependency Preserving Decomposition

- **Informally**: We don't want the original FDs to span two tables.
- R has a dependency-preserving decomposition to X, Y if $F^+ = (F_x \cup F_y)^+$
- **Definition:** F_X all those FDs in F which only mention attrs from X
- Note: F is not necessarily = $F_x \cup F_y$
- Example: R (sailor, boat, date) F: {D → S, D → B}

- Consider decomp. to X (sailor, boat) and Y (boat, date)
- We have F_Y: {D → B}, F_X: {}. $\{D \rightarrow S, D \rightarrow B\}^+ \neq \{D \rightarrow B\}^+$
- To enforce D → S, must join X and Y (expensive)

The above is not dependency preserving

Boyce-Codd Normal Form (BCNF)

- Rel. R with FDs F is in BCNF if, for all $X \rightarrow A$ in F^+
  - $A \subseteq X$ (trivial FD), or
  - X is a super key

X:subset of attributes

A: single attribute

i.e. all non-trivial FDs over R are due to keys.

⇒No redundancy (at least none due to FDs)⇒Most desirable norm. form

Example: The following set of relations are all in BCNF given the following FDs:

Supplier ID → Supplier Name

Supplier ID	Supplier Name
1	Acme
2	Beanery

Item → Desc

Item	Desc
Dynamite	boom
Paint	blue
Flowers	pink

Supplier ID, Item → Price

Supp ID	Item	Price
1	Dynamite	$7
1	Paint	$10
1	Flowers	$3
2	Dynamite	$8

3NF

- Relation R with FDs F is in 3NF if, for all $X \rightarrow A$ in F^+
  - $A \subseteq X$ (trivial dependency) or
  - X is a super key or
  - A is part of some (minimal) key for R **(prime attribute)**

  Minimality of a key (i.e. not a super key) is crucial!

X:subset of attributes

A: single attribute

- BCNF implies 3NF, but 3NF does not imply BCNF

Lossless-join, dependency-preserving decomposition of R into a collection of 3NF relations is always possible.

3NF: Example

- Consider Reserves(Sailor, Boat, Date, CreditCard)
  - F={SBD -> SBDC, S -> C} (not 3NF)

    Why? SBD is the only key, S not a key, C not a key
  - If **additionally** C -> S, then CBD -> SBDC (i.e., CBD is also a key).
    → Now in 3NF!

Note the redundancy in (S, C); 3NF permits this type of redundancy

- Compromise used when BCNF not achievable, or performance considerations

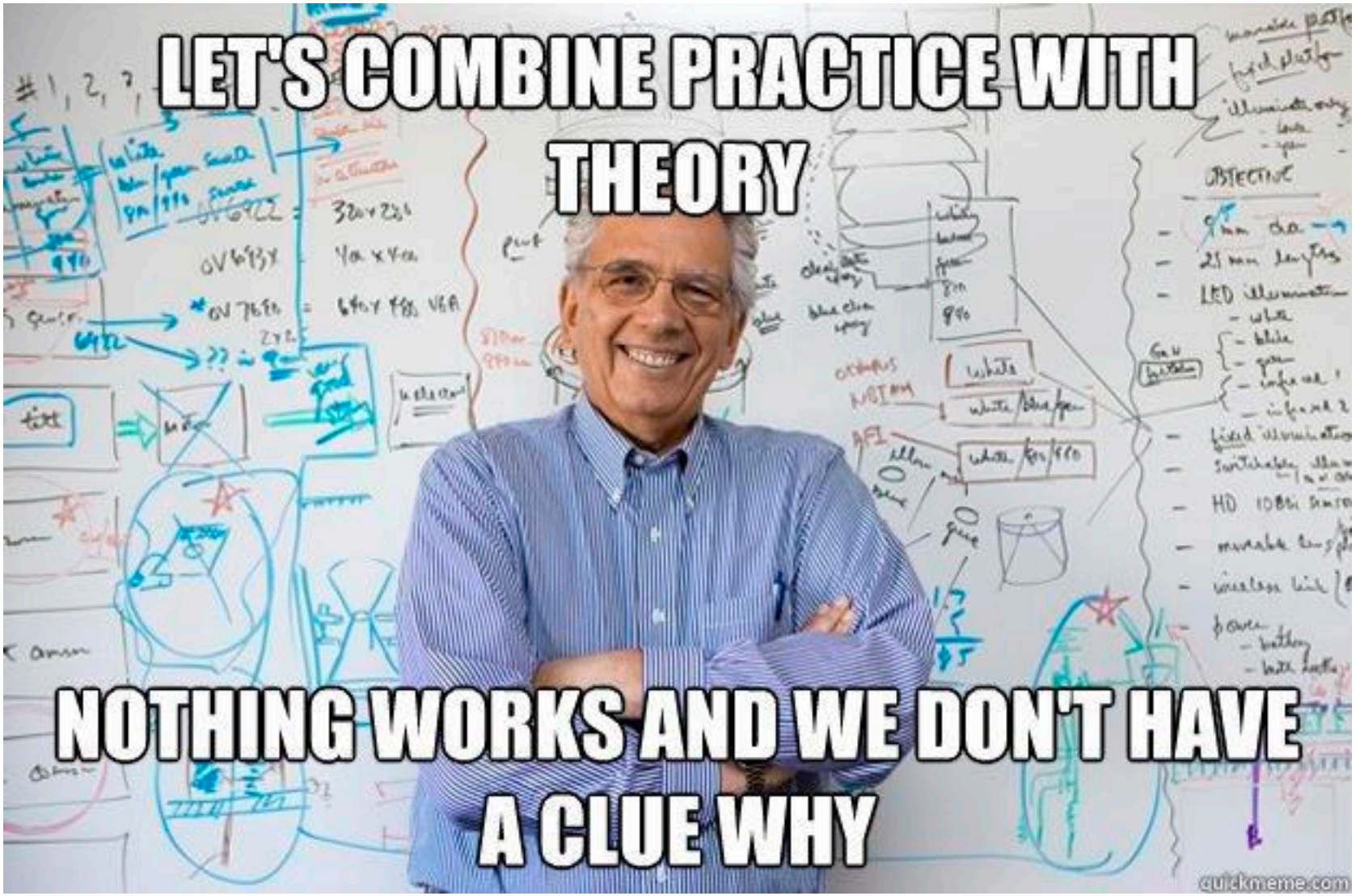
S	B	D	C
S1	B1	D1	C1
S1	B2	D2	C1
S2	B3	D3	C2

Relation R with FDs F is in 3NF if, for all $X \rightarrow A$ in F^+

- $A \subseteq X$ (trivial dependency) or
- X is a super key or
- A is part of some (minimal) key for R **(prime attribute)**

Minimality of a key (i.e, not a super key) is crucial!

Time to practice ☺



Exercise 1: BCNF or 3NF?

- Relation R=(A,B,C,D,E)
- FDs:

  $A \rightarrow BC$

  $CD \rightarrow E$

  $B \rightarrow D$

  $E \rightarrow A$

Hint:
1. Use Armstrong's Axioms to compute F+
2. Identify the keys

- Is R in BCNF?
- Is R in 3NF?

- Reflexivity: If $Y \subseteq X$, then $X \rightarrow Y$ (trivial dependency)
- Augmentation: If $X \rightarrow Y$, then $XZ \rightarrow YZ$ for any Z
- Transitivity: If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$

Exercise 2: FDs & Normal Forms

Suppose you are given the following relation R: ABCDEF

BC -> D

CD -> B

D -> E

ACD -> F

1. Find the keys of R
2. List all of the above FDs that violate BCNF
3. List all of the above FDs that violate 3NF

Decomposition into BCNF

High-Level Algorithm

Input: a relation R with FDs F

1. Identify if any FDs violate BCNF
   - If $X \rightarrow Y$ violates BCNF, decompose R into **R - Y and XY**
2. Repeat for every $X \rightarrow Y$ that violates BCNF.

Output: a collection of relations that are in BCNF

- Does this algorithm provide a lossless join decomposition?
  - Yes! Notice that X is a key for the relation XY
- Is this algorithm dependency preserving? Not necessarily!
- Several dependencies may cause violation of BCNF.
  - The order in which we "deal with" them may lead to different relations!

Algorithm for BCNF
Input: Relation R, FDs F

done = false;

result = {R};

compute F^+;

while (not done) do

  if $\exists$ $R_i \in$ result and R_i is not in BCNF

    let $\alpha \rightarrow \beta$ be a nontrivial FD that holds in R_i

      such that $(\alpha \rightarrow R_i) \notin F^+$ and $\alpha \cap \beta = \varnothing$;

    result=(result$-R_i$) $\cup$ ($R_i - \beta$) $\cup$ (α,β) ;

  else done=true ;

Exercise 3: Fix R to be in BCNF

- Relation R=(A,B,C,D,E)
- FDs: $A \rightarrow BC$, $CD \rightarrow E$, $B \rightarrow D$, $E \rightarrow A$
- Keys: A, BC, CD, E
- $B \rightarrow D$ violates BCNF
- Decompose R:
- Is this decomposition lossless join?
- Is it dependency preserving?

Exercise 4

Suppose you are given the following relation R: ABCDEF with the following FDS:

- BC -> D
- CD -> B
- D -> E
- ACD -> F

Now decompose R into R1 and R2, and specify which decomposition (if any) satisfies lossless join property? Which one is dependency preserving?

- R1: ACDF, R2: ABCDE
- R1: BCD, R2: ABEF

Refining an Initial Schema

- Suppose you are given the following schema. You are asked to normalize it:
- IS (<u>item</u>, name, desc, loc, price)
  S (<u>name</u>, addr)

- Requirement: A supplier keeps all items of the same name in the same location FD: name → loc

Normalization Example

IS (<u>item</u>, name, desc, loc, price)
S (<u>name</u>, addr)

FDs = {i → ndlp, n → la}

- IS is not in BCNF, due to n → l
- Break it up: IS(<u>i</u>,n,d,p), Loc(<u>n</u>,l)
- S(<u>n</u>,a) remains unchanged
- Now notice same key for S and Loc, so merge
- Loc (name, addr, loc)

Refining an Initial Schema

- IS (item, name, desc, loc, price)
  S (name, addr)

- A supplier keeps all items of the same name in the same location FD: name → loc

```
Solution:
IS (item, name, desc, price)
Loc (name, addr, loc)
```

New Schema

Normalization: Summary

- Bad schemas lead to redundancy
  - Redundant storage, update, insert, and delete anomaly
- To "correct" bad schemas: decompose relations
  - Must be a lossless-join decomposition
  - Would like dependency preserving decompositions
- Desired Normal Forms
  - BCNF: allow only super-key functional dependencies
  - 3NF: allow dependencies with prime attributes on the RHS
    - Allows a limited form of redundancy
    - Trades off performance (avoid joins) for redundancy

Exercises

19.1, 19.5, 19.25

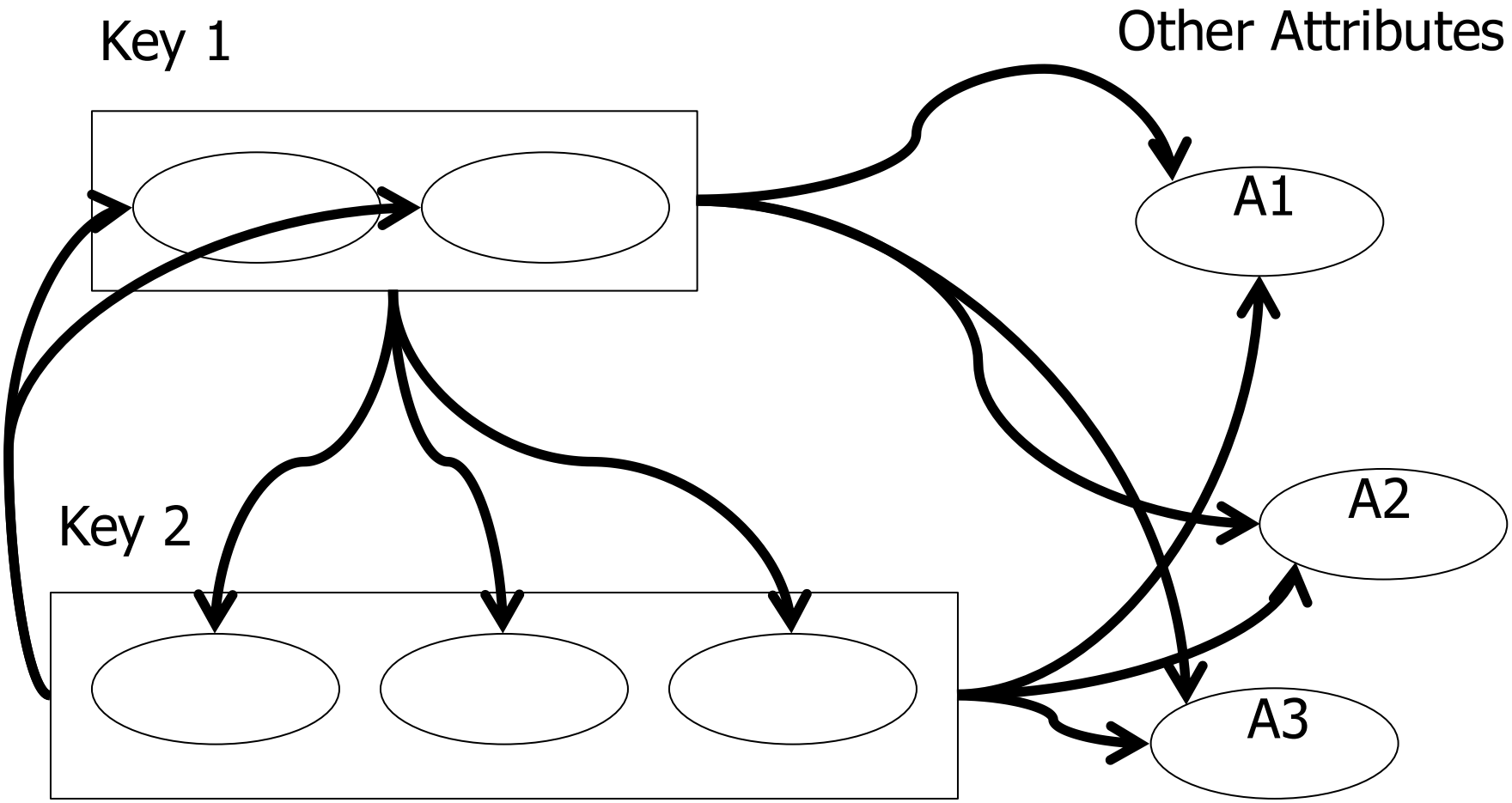
Next…

Storage and Indexing

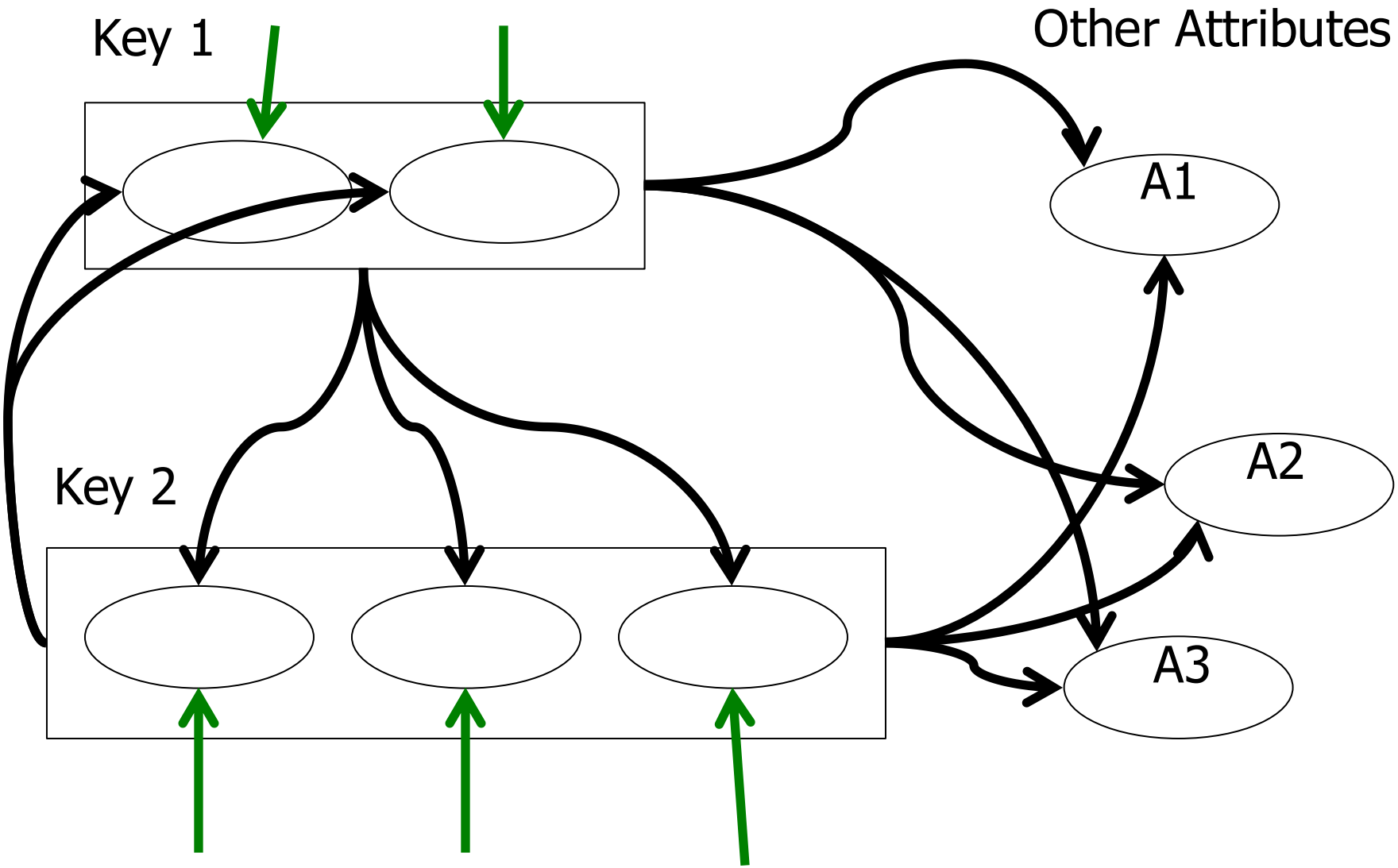
Optional Slides

- The following slides are optional
- They're just a visual aid that Mozafari came up with when he took the undergrad DB class himself (2004)
  - Thankfully, normal forms haven't changed since!
- They helped him memorize all these confusing normal forms when he was preparing for his own exam
  - Back then, they had to learn about 4NF and 5NF too!
  - Lucky you: don't even have to learn about 2NF these days!
- Use these illustrations if you find them useful

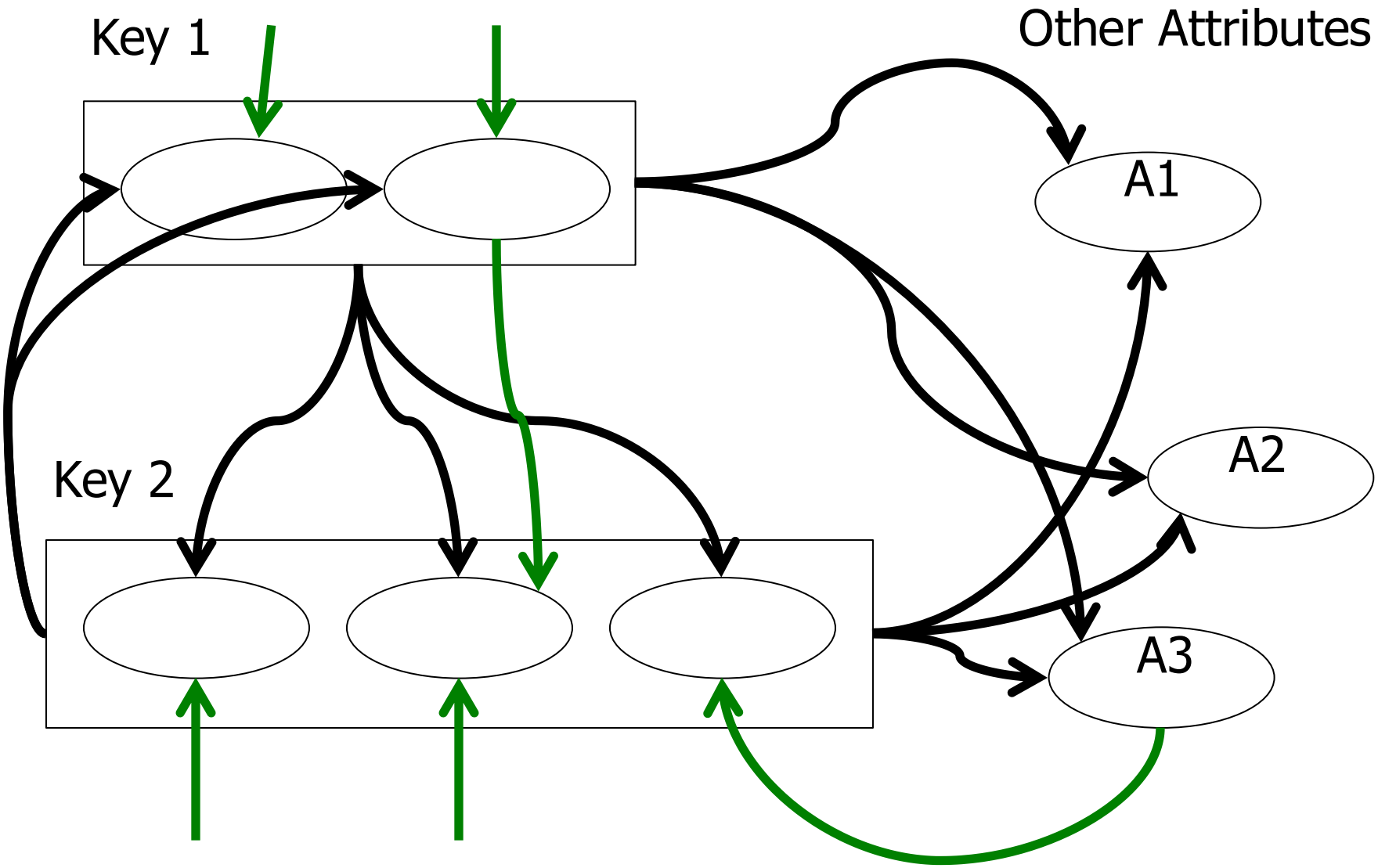
BCNF



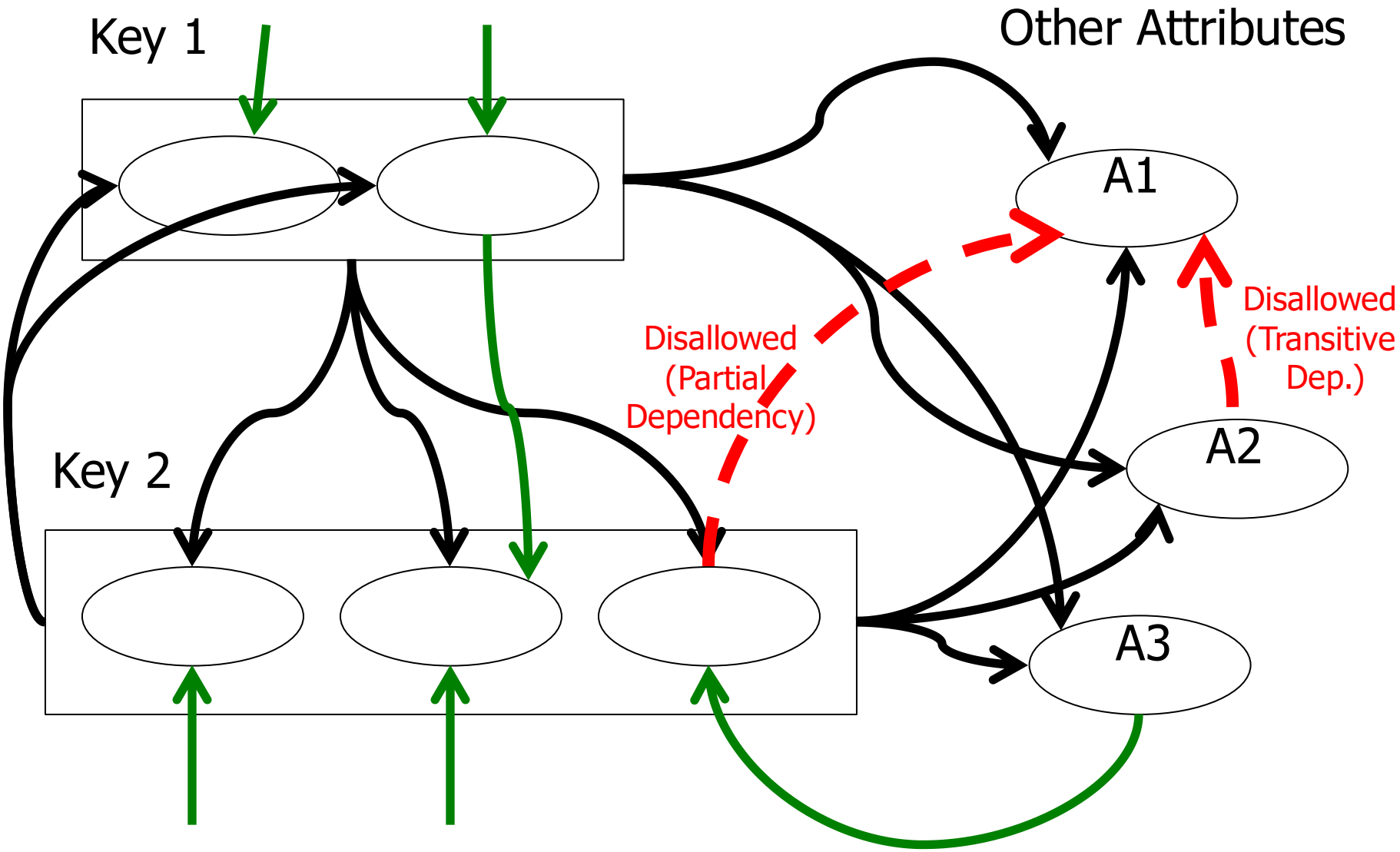
3NF



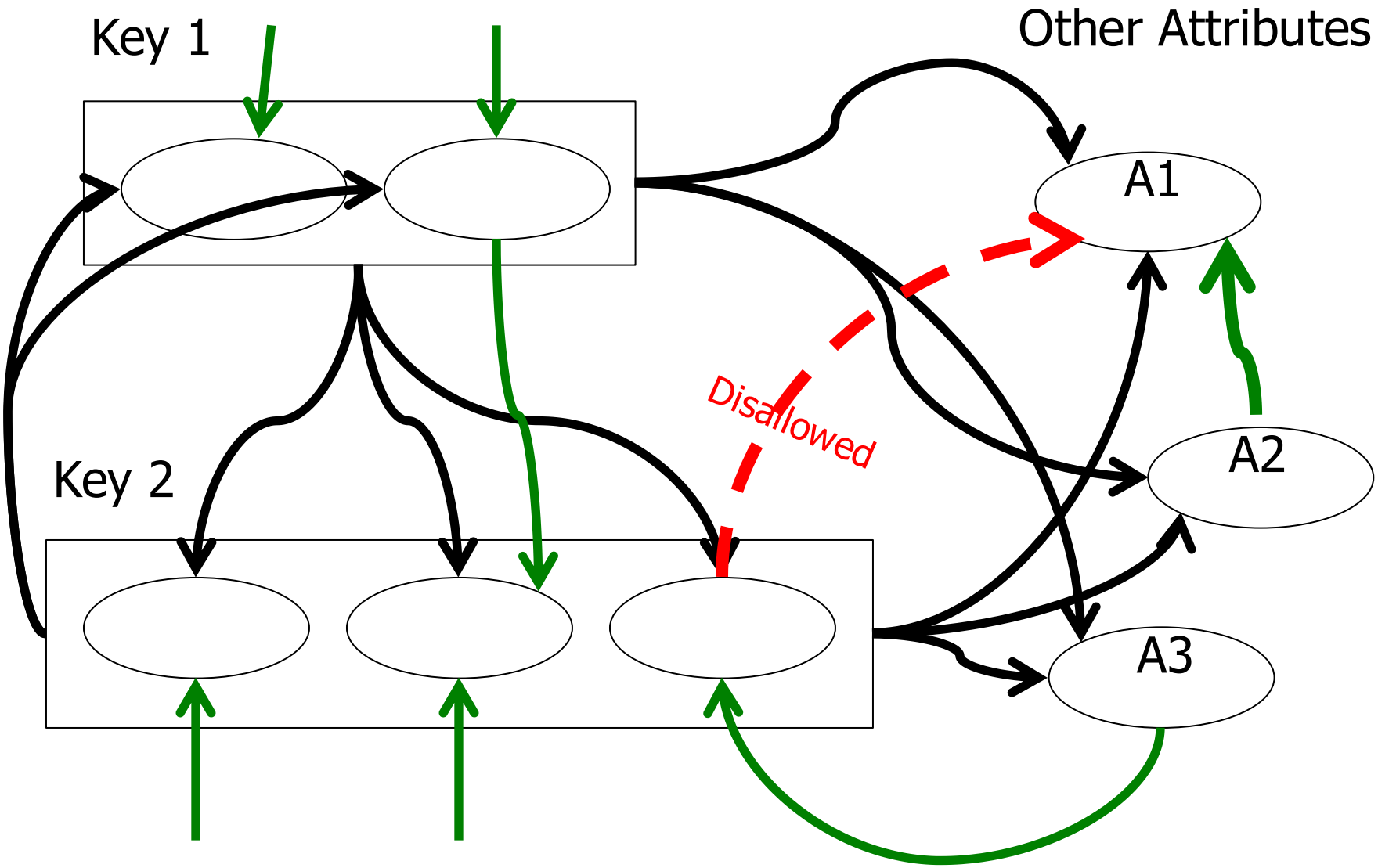
3NF



3NF


Key 1
Other Attributes
A1
Disallowed
(Transitive
Dep.)
Disallowed
(Partial
Dependency)
A2
Key 2
A3

2NF



Basic Normalization

- Write keys -> attribute mappings
- One table for each

Supplier ID	Supplier Name	Item	Desc	Price
1	Acme	Dynamite	boom	$7
1	Acme	Paint	blue	$10
1	Acme	Flowers	pink	$3
2	Beanery	Dynamite	boom	$8
2	Beanery	Dynamite	boom	$8

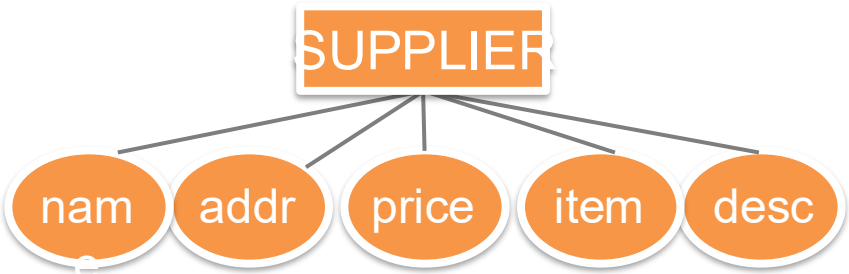
- Supplier ID → Supplier Name
- Item → Desc
- Supplier ID, Item → Price

Supplier ID	Supplier Name
1	Acme
2	Beanery

Item	Desc
Dynamite	boom
Paint	blue
Flowers	pink

Supp ID	Item	Price
1	Dynamite	$7
1	Paint	$10
1	Flowers	$3
2	Dynamite	$8

Example: Constraints on Entity Set



- S(name, item, desc, addr, price)
  - FD: {n,i} → {n,i,d,a,p}
  - FD: {n} → {a}
  - FD: {i} → {d}
- Decompose to: NA, ID, INP

- Spl(name, item, price)
  - FD: {n,i} → {n, i, p}
- Sup(name, addr)
  - FD: {n} → {n, a}
- Item (item, desc)
  - FD: {i} → {i, d}

ER design is subjective and can have many E + Rs
FDs: deeper understanding of schema